Journal of Applied Sport Science Research
1987, Volume 1, Number 3, pp. 48-52

Metabolic Cost of a Preparatory Phase of Training in Weight Lifting: A Practical Observation

Dwight Scala, Jim McMillan, Danny Blessing,
Ralph Rozenek and Mike Stone
National Strength Research Center
Department of Health, Physical Education
and Recreation
Auburn University, Alabama 36849

ABSTRACT

The metabolic cost of weight training, especially advanced programs, has been poorly researched and is poorly understood. This study describes the metabolic, heart rate, and blood pressure responses of three weight lifters during a week of preparation phase training. Oxygen uptake ($\dot{V}O_2$) was measured with a Beckman MMC, resting heart rate (RHR) by palpitation, and resting blood pressure (RBP) by auscultation. Twelve workouts were performed during the week. The average caloric expenditure (L O_2 X 5 + Kcal) was 9.4Kcal · min^{-1} and 3918 Kcal/wk. Larger muscle mass exercises (i.e., squats, pulls, etc.) averaged 11.5 Kcal · min^{-1}, and small muscle mass exercises (i.e., bench press, sit-ups, etc.) averaged 6.8 Kcal · min^{-1}. Resting HR and RBP were largely unaffected. It appears that the volume and intensity of this type of training is sufficient to elicit beneficial alterations in body composition, serum lipids and possible cardiovascular function.

KEY WORDS: Weight training, heart rate, blood pressure, oxygen uptake, caloric expenditure.

Introduction

Weight training has become an integral part of the training programs of many types of athletes. Coaches and athletes recognize the important contributions that increased muscular strength and power can make toward improved performance. However, weight training programs are often implemented with little knowledge of the energy cost of weight training.

Noncircuit weight training has been criticized as providing minimal or no stimulus to improve aerobic functions or reduce body fat (17, 16). However, it should be noted that highly weight trained athletes do possess higher than average max $\dot{V}O_2$'s ($\bar{X} \cong$ 55 ml · kg^{-1} · min^{-1}) and low percent body fat ($\bar{X}$ + 10%) (4, 20, 21). Recently, longitudinal studies using short-term noncircuit weight training programs have produced small but significant increases in max $\dot{V}O_2$ both L · min^{-1} (8, 9, 24), and ml · kg^{1} · min^{-1} (24) and substantial significant losses in body fat (23, 24). Furthermore, total caloric cost may be associated with beneficial alterations in blood lipids (14, 29). Noncircuit weight training has been shown to produce positive changes in the serum lipid levels of middle-aged sedentary men (12, 7). These studies suggest that noncircuit weight training could have a large metabolic turnover rate of energy expenditure. If the metabolic turnover is high and/or it produces a high total energy cost, this could have considerable importance in the use of weight training as part of a physical fitness program or as part of various athletic training programs. Furthermore, the addition of weight training producing a high total energy cost could result in chronic fatigue and a decrease in performance (overtraining) if not properly integrated with other training procedures.

Typically, anaerobic exercise produces a large O_2 debt relative to the time spent exercising (16). Weight training (anaerobic exercise) using moderate to high repetitions, multiple sets, large muscle mass exercises, and especially if using multiple exercise sessions per day, may produce a very large total energy cost, most of which may appear as O_2 debt. Many athletes use this type of extensive weight training program at various times during their training year (e.g., shot putters, discus throwers, power lifters, etc.). It is commonly used by weight lifters during parts of their preparation phase (24, 25).

The purpose of this study was to estimate the energy cost of

Table 1. Physical and Performance Variables of the Subjects at the Time of the Study

Subject	Best Snatch (kg)	Best C&J (kg)	BW (kg)	Max $\dot{V}O_2$ (L · min^{-1})	Max $\dot{V}O_2$ (ml · kg^{-1} · min^{-1})	Age (Years)
1	85.0	100.0	80.0	3.88	48.6	32
2	95.0	110.0	80.0	3.68	44.9	25
3	115.0	150.0	108.0	4.33	40.9	34
$\bar{X}$ (N + 3)	98.3	120.0	89.3	3.96	44.8	31

Average values for college-aged males (untrained) from this laboratory.
Age — 20 years
BW — 77 kg
Max $\dot{V}O_2$ (L · min^{-1}) - 3.05
Max $\dot{V}O_2$ (ml · kg^{-1} · min^{-1}) - 39.6

% Difference from average
Age — +55 years
BW — +15.9
Max $\dot{V}O_2$ (L · min^{-1}) - +29.8
Max $\dot{V}O_2$ (ml · kg^{-1} · min^{-1}) - +13.1

a weight training program similar to that used by many olympic style weight lifters during their preparation phase. The energy cost was estimated by determining exercise and recovery oxygen consumption values during a one-week training period. Additional measures of the intensity of training included resting HR and blood pressure.

Methods

Three weight trained athletes, after giving informed consent, served as subjects. Subject 3 was a former national class weight lifter, recently returned to training after a six-month layoff. Subject 2 was a former power lifter training for weight lifting competition. Subject 1 was a novice weight lifter. The subjects' physical and performance characteristics at the time of the study are shown in **Table 1**. None of the subjects reported taking anabolic steroids at the time of the study.

Maximum $\dot{V}O_2$ was determined by using a Beckman metabolic measurement cart (MMC) and a Monark cycle ergometer. The $\dot{V}O_2$ exercise protocol began at 9 watts (1st minute) and was increased by 65 watts each minute until 195 watts was reached. Thereafter it was increased by 32.5 watts until exhaustion. The pedal rate was 70 rpm's. Maximum $\dot{V}O_2$ was measured one week prior to the experiment.

Oxygen consumption during weight training sessions was measured using the MMC. Resting $\dot{V}O_2$ was considered as an average of three consecutive minutes, after lying down in a darkened room for 15 minutes. After a brief warm-up, $\dot{V}O_2$ of each set and each rest period of each exercise was measured. This was accomplished by manually stopping and starting the MMC. The time ($\dot{V}_E$, etc.) lost by stopping and starting the MMC, amounts to only two to four seconds and is negligible. This method of $\dot{V}O_2$ sampling allows a more accurate estimate of exercise $\dot{V}o_2$ versus resting $\dot{V}O_2$ and allows the subjects to take *normal* rest intervals between sets. Recovery $\dot{V}O_2$ was measured for 10 minutes following the end of an exercise session. Caloric cost was estimated using 5 Kcal per liter of O_2. Further indications of training stress (11) were measures of resting heart rate (RHR) and blood pressure (RBP). Resting heart rate before each exercise session was measured by radial artery palpation and RBP by auscultation after 15 minutes of lying quietly in a darkened room.

The week's training schedule is shown in **Table 2**. Percen-

Table 2. Daily Training Protocol.

Monday A.M.$_1$
1. Squats
2. Vertical jumps (hold dumbbells)
3. Leg curls

Monday A.M.$_2$
1. Squats (narrow stance)
2. Back ¼ squats (power rack)
3. Vertical jumps
4. Straight legged dead lifts

Monday P.M.
1. Dumbbell bench press
2. Seated dumbbell press
3. Lateral raises
4. Pulldowns (lat. machine)
5. Sit-ups (crunches)

Tuesday A.M.
1. Snatch pulls (floor)
2. Snatch pulls (thigh)
3. Snatch grip shoulder shrugs

Tuesday P.M.
1. Snatch pulls (knee)
2. Snatch grip shoulder shrugs
3. Vertical jumps
4. Hyperextensions
5. Dumbbell inclines
6. Lying laterals (prone)
7. Sit-ups (crunches)

Wednesday
1. Triceps work
2. Sit-ups (crunches)

Thursday A.M.$_1$
1. Dead stop squats (power rack)
2. Clean pulls (floor)
3. Clean grip shoulder shrugs
4. Vertical jumps

Thursday A.M.$_2$
1. Pulls (floor)
2. Pulls (thigh)
3. Clean grip shoulder shrugs

Thursday P.M.
1. Vertical jumps (deep)
2. Hyperextensions
3. Dumbbell bench press
4. Pulldowns (lat. machine)
5. Sit-up (crunches)

Friday
1. Squats (narrow stance) — light
2. Sit-ups (crunches)

Saturday
1. Clean grip pulls (thigh)
2. Clean grip shoulder shrugs
3. Vertical jumps (holding dumbbells)
4. Hyperextensions
5. Inclines
6. Seated dumbbell press
7. Lateral Raises
8. Sit-ups (crunches)

Each session except Wednesday and Monday P.M. was preceded by stretching and power snatches with 40kg 3 x 10. (Power snatches are included in the caloric cost of the session.)

Major exercises were preceded by warmup sets (10 reps.) with set percentages of maximum for the lift. Target weights for major exercises were 62.5 percent of maximum for 3 x 10. Mean time between sets was approximately 2.3 minutes.

Table 3. Caloric Cost per Training Session and Weekly Means.

	Monday			Tuesday		Wednesday	Thursday			Friday	Saturday	$\bar{X}$ (Excl. Wed.)
	1	2	3	1	2	1	1	2	3	1	1	
$\bar{X}$ Kcal Stretch	21.9	16.4	16.0	19.4	21.4	11.6	20.7	19.0	22.8	22.6	31.4	21.3
$\bar{X}$ Kcal min^{1} Stretch	4.7	4.4	4.0	4.1	4.5	3.5	4.4	4.2	4.6	3.8	4.1	4.3
$\bar{X}$ Kcal Workout	368.5	341.7	272.7	379.7	442.1	30.6	409.7	339.6	206.5	158.9	374.2	332.1
$\bar{X}$ Kcal · min^{1} Workout	11.0	8.9	6.8	10.3	9.1	4.1	11.2	10.4	7.8	9.7	8.6	9.4
$\bar{X}$ Kcal Recovery	59.4	36.8	26.6	42.4	33.0	20.5	54.0	46.3	28.1	30.5	27.7	38.5
$\bar{X}$ Kcal · min^{1} Recovery	5.9	3.7	2.7	4.2	3.3	2.1	5.4	4.6	2.8	3.1	2.8	3.9
$\bar{X}$ Kcal Total	449.8	394.9	315.3	441.5	496.5	62.7	484.4	405.8	257.4	239.0	433.3	391.8
$\bar{X}$ Kcal · min^{1} Total	8.8	7.6	5.8	8.5	7.8	3.0	9.4	8.6	6.2	6.8	7.1	7.7

tages for major exercises were based on each subject's best lifts at the time of the study. This training schedule is indicative of the preparation phase of training used by various strength-power athletes (23, 24). The experiment was carried out during the third week of the subjects' preparation phase (third week of this type of training).

Results

The energy cost during the third week of preparation phase training (see **Table 2**) was determined for three moderately trained weight lifters. The average energy cost per training session is shown in **Table 3**. The average total Kcal used per week during training was 3,918. The net energy cost (total-resting) per week was about 3,500 Kcal. The following measures (*excluding Wednesday*) suggest a typical exercise session using this type of training produced an average rate of caloric use of 4.3 Kcal · min for warm-up, 9.4 Kcal · min^{-1} for the workout and 3.9 Kcal · min^{-1} for ten minutes of recovery.

Table 4 presents $\dot{V}O_2$ and caloric cost for large-vs-small muscle mass exercises (i.e., squats, pulling movements, etc.). The rate of caloric cost (11.5 vs. 6.8 Kcal · min^{-1}) and total (2705 vs. 861 Kcal) caloric cost was considerably higher for large muscle mass exercises. An average exercise session lasted 36.2 minutes; of this, large muscle mass exercises averaged 23.5 minutes (65 percent) and small muscle mass 12.7 minutes (35 percent). Average $\dot{V}O_2$ for the large muscle mass exercises represented 58 percent of the subject's maximum $\dot{V}O_2$ (ml · Kg · min^{-1}) and 34 percent for small muscle mass exercises.

In general, rest intervals produced higher $\dot{V}O_2$'s than did exercise. This was found consistently in all training sessions. Recovery was not complete after 10 minutes post exercise. Again, this was consistent across the week. Supporting this is the observation that resting $\dot{V}O_2$ for the week average 294.3 ml · min^{-1} and average $\dot{V}O_2$ for the tenth minute of recovery was 532.3 ml · min^{-1} (**Figure 1**).

Resting heart rate tended to rise during multiple session days and throughout the week resting $\dot{V}O_2$ and blood pressure showed inconsistent changes. None of the subjects displayed hypertension. Mean heart rates and blood pressures were below typical averages (**Table 5**).

Table 4. $\dot{V}O_2$ and Energy Cost of Large *vs.* Small Muscle Mass Exercises

	Small Muscle $\bar{X}$	Large Muscle $\bar{X}$
$\dot{V}O_2$ L · Min^{-1}	1.36	2.30
$\dot{V}O_2$ Ml · K · Min^{-1}	15.3	25.8
% Max $\dot{V}O_2$ L · Min^{-1}	34.3	58.1
% Max $\dot{V}O_2$ Ml · Min^{-1} · Kg^{-1}	34.1	57.6
O_2 Consumed (L)	172.27	540.98
Kcal Expended	861.4	2704.9
Total Time (Min.)	126.81	234.9
Kcal · Min	6.8	11.5

Figure 1. Average of 10 Minutes of Recovery Post Exercise.

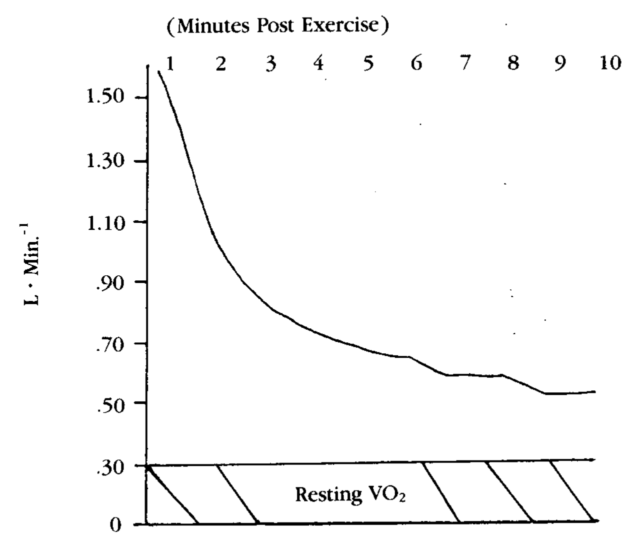

Table 5. Heart Rates, Blood Pressure, and $\dot{V}O_2$ for Each Training Session

		Monday			Tuesday		Wednesday	Thursday			Friday	Saturday
Resting		1	2	3	1	2	1	1	2	3	1	1
Heart Rates	S1	59	61	78	66	77	72	65	71	60	66	67
	S2	62	66	68	64	72	60	60	64	72	60	61
(beats ·	S3	54	60	63	66	77	50	56	60	62	54	55
min^{-1})	$\bar{X}$	58	62	70	65	75	61	60	65	65	60	61
Blood	S1	120/69	106/68	105/60	116/80	108/70	110/62	109/64	121/80	108/65	110/62	109/65
Pressure	S2	108/79	112/70	118/70	111/70	111/77	116/78	109/63	104/66	124/60	120/70	105/65
(mm · Hg)	S3	134/88	132/90	126/70	116/80	108/70	131/81	147/85	134/80	142/75	141/81	128/70
	$\bar{X}$	121/76	117/76	116/67	114/77	109/72	119/74	122/71	120/75	125/67	127/71	114/67
$\dot{V}O_2$	S1	265	280	290	310	333	276	277	280	337	313	290
(M1 ·	S2	300	293	263	186	280	297	345	275	357	285	290
min^{-1})	S3	397	417	285	263	293	290	277	340	213	300	227
	$\bar{X}$	321	330	279	253	302	288	300	332	302	299	269

Tables 6a, b, and c show the relationship of exercise $\dot{V}O_2$ to rest $\dot{V}O_2$. The majority of O_2 cost and, therefore, energy cost, occurs during rest.

Discussion

A typical workout lasting about 36.2 minutes used about 330 Kcal at a rate of 9.4 Kcal · min^{-1} (**Table 3**). Using the method of Wilmore et al. (30) and combining the recovery energy with the exercise session produces a rate of 10.6 Kcal · min^{-1}. This corresponds qualitatively to very heavy exercise (2, 18, 28).

In no case (*excluding Wednesday*) was recovery complete after 10 minutes. The recovery $\dot{V}O_2$ of very intense exercise may persist above resting levels for a considerable time after work is completed (22, 15). Thus, the final total caloric cost was likely above the total values presented (see **Figure 1**).

A major portion of each exercise session is devoted to large muscle mass exercises (e.g., squats, pulls, jumping, etc.). These types of exercises clearly produce a higher energy cost (**Table 4**). Furthermore, they are placed (consecutively) at the beginning of the exercise session preceding the small muscle mass exercises. This creates about 15 minutes to 30 minutes of relatively high heart rates (22), high $\dot{V}O_2$'s and average about 11.5 Kcal · min^{-1}. Some studies have suggested that exercise $\dot{V}O_2$'s as low as 45 percent to 50 percent are sufficient to increase maximum $\dot{V}O_2$, provided they are maintained long enough (6). These large muscle mass exercises produced mean exercise $\dot{V}O_2$'s of 58.1 percent of maximum $\dot{V}O_2$ L · min^{-1} and 57.6 percent ml · Kg^{-1} · min^{-1}. Thus, there is a possibility of an aerobic training effect. Using a similar training protocol, Stone et al. (24) found small but significant increases in maximum $\dot{V}O_2$ (both absolute and relative) in previously untrained young men over seven weeks.

Small muscle mass exercises are not as important from an energy production/cost standpoint. However, placement of the smaller muscle mass exercises at the end of a training session may provide a warm-down. This can be important in making a smoother transition toward resting metabolic levels after intense intermittent exercise (5).

The subjects did not feel as though they were completely recovered between workouts on multiple session days. Furthermore, the subjects reported a subjective feeling of increasing fatigue from Monday to Wednesday and from Thursday through Saturday. Resting $\dot{V}O_2$, blood pressure, and heart rate are known to rise with chronic fatigue and overtraining. The changes observed in resting blood pressure or $\dot{V}O_2$ were not indicative of incomplete recovery. However, heart rate did rise five to twelve beats · min^{-1} from Monday to Wednesday and slightly (three to seven beats · min^{-1}) from Thursday to Saturday (**Table 5**). Perhaps these increases in heart rate reflect fatigue. Previous experience with this type of high volume training has subjectively shown it to be very fatiguing (much being localized). Thus, sufficient recovery periods must be present through a week's training; this is the purpose of the light Wednesday training session, single session training on Friday and Saturday, and no training on Sunday.

The generally higher resting $\dot{V}O_2$'s (L · min^{-1}) (between sets) relative to exercise $\dot{V}O_2$, especially during the first minute, (**Table 6a, b, and c**) are likely due to a number of factors including: (1) some breath holding occurs during lifting, thus reducing $\dot{V}_E$; (2) increased blood lactate stimulating chemoreceptors (pH), and increasing $\dot{V}_E$ during the rest; (3) increased interstitial [K+] stimulation of chemoreceptors causing an increased $\dot{V}_E$ at rest; and (4) other metabolic stimulators (e.g., increased temperature, heart rate, hormonal influences, etc.) (1, 17). Similar arguments can be made for the recovery O_2 debt.

Practical Applications

Previous investigations of weight training energy cost have used primarily small muscle mass exercises and/or relatively low training intensities (16, 30, 10). These investigations found the $\dot{V}O_2$ and estimated total energy cost of weight training to be smaller than in the present investigation. The present observation shows that the extensive use of large muscle mass exercises coupled with high training volumes (repetitions) not only produces a relatively high $\dot{V}O_2$ and metabolic rate but also a very high total caloric cost. This suggests that there is a reasonable probability of beneficially altering aerobic power and especially body composition. It further suggests that similar training programs would meet many of the basic requirements for the preparation of strength-power athletes (24). Thus, in the athletic or non-athletic setting considerable thought should be given to

Table 6a. Exercise vs Rest O_2 for Monday-Wednesday $\bar{X}$ (L)

	Monday			Tuesday		Wednesday
	1	2	3	1	2	1
Work	14.8	16.3	14.4	15.2	20.5	2.6
Rest	58.9	52.1	39.9	60.7	68.0	3.5
Total	73.7	68.4	54.3	75.9	88.5	6.1

Table 6b. Exercise vs Rest O_2 for Thursday-Saturday $\bar{X}$ (L)

	Thursday			Friday		Saturday	$\bar{X}$ Session
	1	2	3	1	2	1	
Work	22.7	14.4	13.0	15.2	20.5	2.6	15.3
Rest	59.3	53.5	28.3	60.7	68.0	3.5	46.0
Total	82.0	67.9	41.3	75.9	88.5	6.1	61.3

Table 6c. Exercise vs Rest for O_2 per Day $\bar{X}$ (L)

	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	Total $\bar{X}$ O_2 for Week
Work	45.5	35.7	2.6	50.2	11.4	23.0	168.3
Rest	150.9	128.7	3.5	141.1	25.8	56.1	506.1
Total	196.4	164.4	6.1	191.3	37.2	79.1	674.4

including extensive use of large muscle mass, multi-segment exercises coupled with high repetitions during appropriate periods of training.

Residual fatigue from weight training (both general and local) must be considered in extensive weight training or combined training programs. Empirically high volume lower intensity weight training (large total energy expenditure) is associated with chronic fatigue to a much greater extent than lower volume high intensity training. Part of this chronic fatigue may result from chronic depression of glycogen stores as a result of repeated large energy expenditures (3, 6). Glycogen depletion is associated with decreased strength and endurance (13, 19). Because of the high total energy cost, multi-session training (e.g., weight training and running and/or sports practice) at too high an intensity and volume may quickly produce overwork resulting in decreased performance, chronic fatigue, and other overtraining symptoms (26). The weight training program used produced a considerable total energy cost. The rate of caloric expenditure ranges from moderate to very heavy and compares well with many primarily aerobic exercises.

References

1. Brooks, G. A., Hittelman, K. J., Faulkner, J. A., and Beyer, R. E. (1971) Temperature, skeletal muscle mitochondrial functions and oxygen debt. **Am. J. Physiol.** 220:1053-1068.
2. Broucha, L., and Radford, E. (1960) The cardiovascular system in muscular activity. In: W. Johnson (ed.), **Science and Medicine of Exercise and Sports**. New York, McGraw-Hill.
3. Costill, D. L., et al. (1980) Nutrition for endurance sport: Carbohydrate and fluid balance. **Int. J. Sports Med.** 1:2-14.
4. Fahey, T., Akka, L., and Rolph, R. (1975) Body composition and $\dot{V}O_2$ max of exceptionally weight trained athletes. **J. Appl. Physiol.** 39:559-561.
5. Fox, E. L., and Mathews, D. K. (1981) **The Physiological Basis of Physical Education and Athletics** (3rd ed.). New York, W. B. Saunders.
6. Gaesser, G. A., and Rich, R. G. (1984) Effects of high and low-intensity exercise training on aerobic capacity and blood lipds. **Med. Sci. Sports Exerc.** 16:269-274.
7. Goldberg, L., Elliot, D. L., Schutz, R. W., and Kloster, F. E. (1984) Changes in lipid and lipoprotein levels after weight training. **JAMA** 252:504-506.
8. Hickson, R. C. (1980) Interference of strength development by simultaneously training for strength and endurance. **Eur. J. Appl. Physiol.** 215:255-263.
9. Hickson, R. C., Rosen-Koetter, M. A., and Brown, M. M. (1980) Strength training effects on aerobic power and short-term endurance. **Med. Sci. Sports Exerc.** 12:336-339.
10. Hickson, J. F., Buono, M. J., Wilmore, J. H., and Constable, S. H. (1984) Energy cost of weight training exercise. **Nat. Strength and Conditioning Assoc. J.** 6(5):22-66.
11. Hunter, G. R., and McCarty, J. P. (1983) Pressor response associated with high-intesity anaerobic training. **Physician Sportsmed.** 11:151-162.
12. Johnson, C. C., Stone, M. H., Lopez-S., A., Hebert, J. A., Kilgore, L. T., and Byrd, R. J. (1982) Diet and exercise in middle-aged men. **J. Amer. Diet. Assoc.** 81:695-701.
13. Karlsson, J., Sjodin, B., Jacobs, I., and Kaiser, P. (1981) Relevance of muscle fibre type to fatigue in short intense and prolonged exercise in man. 1981 Human muscle fatigue: Physiological mechanisms. London, Pitman Medical (Ciba Foundation Symposium 82):52-74.
14. Leon, A. S. (1985) Physical activity levels and coronary heart disease. In L. Goldberg and D. Elliot (Eds.), **The Medical Clinics of North America** 69(1):3-20.
15. Margaria, R., Edwards, H. T., and Dill, D. B. (1933) The possible mechanism of contracting and paying the oxygen debt and the job of lactic acid in muscular contraction. **Am. J. Physiol.** 106:687-714.
16. McArdle, W. D., and Foglia, G. F. (1969) Energy cost and cardiorespiratory stress of isometric and weight training exercises. **J. Sports Med. Phys. Fitness** 9:23-30.
17. McArdle, W. D., Katch, F. I., and Katch, V. L. (1981) **Exercise Physiology**. Philadelphia, Lea and Febiger, 300.
18. Passmore, R., and Durnin, J. V. G. A. (1955) Human energy expenditure. **Physiol. Rev.** 35:801-814.
19. Rozenek, R., and Stone, M. H. (1984) Protein metabolism related to athletics. **Nat'l Strength and Conditioning Assoc. J.** 6(2):42-62.
20. Saltin, B., and Astrand, P. O. (1967) Maximal oxygen uptake in athletes. **J. Appl. Physiol.** 23:353-358.
21. Stone, M. H., Carter, D., Smith, D. P., and Ward, T. (1979) Olympic weight lifting: Physiological characteristics of the athletes. (In J. Terands, ed.) **Science of Weight Lifting**. Del Mar, California, Academic Publishers, 45-53.
22. Stone, M. H., Ward, T., Smith, D. P., and Rush, M. (1979) Olympic weight lifting: Metabolic consequences of a workout. (In J. Terands, ed.) **Science of Weight Lifting**. Del Mar., California, Academic Publishers, 54-67.
23. Stone, M. H., Blessing, D., Byrd, R., Tew, J., Boatwright, D., Karr, B., Johnson, C., Lopez-S, A., Johnson, R., and Carter, DS. (1982) Physiological effects of a short-term resistive training program on middle-aged untrained men. **Nat'l Strength and Conditioning Assoc. J.** 4(5):16-20.
24. Stone, M. H., Wilson, G. D., and Rozenek, R. (1983) Cardiovascular responses to short-term olympic weight training in young men. **Can. J. Appl. Sports Sci.** 8:134-139.
25. Stone, M. H., O'Bryant, H., Garhammer, J., McMillan, J., and Rosenek, R. (1982) A theoretical model of strength training. **Nat'l Strength and Conditioning Assoc. J.** 4(4):36-39.
26. Stone, M. H., and O'Bryant, H. (1985) **Weight Training: A Scientific Approach**. Minneapolis, Burgess.
27. Vorobeyev, A. V. (1978) Weight lifting, Budapest, International Weight Lifting Federation (Translated from Russian by W. J. Brice).
28. Wells, J. B., Balke, B., and Van Fossen, D. (1957) Lactic acid accumulation during work. A suggested standardization of work classification. **J. Appl. Physiol.** 10:51-55.
29. Williams, T. P., Wood, P. D., and Haskell, et al. (1982) The effects of running mileage and duration on plasma lipoprotein levels. **J. Am. Med. Assoc.** 247:2674-2679.
30. Wilmore, J. H., Parr, R. B., Ward, R., Vodak, P. A., Barston, T. J., Pipes, T. V., Grimditch, G., and Leslie, P. (1978) Energy cost of circuit weight training. **Med. Sci. Sport.** 10:75-78.